

Hawkin Dynamics

Beta API Specification Doc

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Before You Start

This API is designed to get data out of your Hawkin Dynamics database into your own database. It is not designed to be accessed from client applications directly. There is a limit on the amount of data that can be returned in a single request. As your database grows, it will be necessary to use the from and to OR the syncFrom and syncTo parameters to limit the size of the responses. Responses that exceed the memory limit will fail. It is advised that you design your client to handle this from the beginning. A recommended pattern would be to have two methods of fetching data. A scheduled pull that uses the syncFrom and syncTo parameters to constrain the returned data to only tests that have occurred since the last fetch e.g. every day or every 5 minutes. And then a pull that fetches the entire database since you began testing that is only executed when necessary. The recommended way of doing this is to generate the from and to parameters for each month since you started and send a request for each either in parallel or sequentially.

This documentation references <https://cloud.hawkindynamics.com/api> for brevity however the host name for your organization may be different. Generally, the host name will depend on your location and/or when you started collecting data with the system.

Americas: <https://cloud.hawkindynamics.com>
Europe: <https://eu.cloud.hawkindynamics.com>
Asia/Pacific: <https://apac.cloud.hawkindynamics.com>

Please reach out to techsupport@hawkindynamics.com and we can confirm the correct host for your organization.

Configuration Options

There are some options we can enable specifically for your organization if you wish. These include:

Rounding of Metrics

By default, the API rounds metrics using the same rounding rules that the Hawkin Capture and Hawkin Cloud Web Application use. We can turn this off if you wish and all metrics are rounded to a maximum of 4 decimals places.

Creation of Athletes

By default, the API will allow the creation of athletes. We can disable this option if you wish

Test Data Structure

By default, the API returns the test data as described below, an object with the metrics and the test metadata at the same level. We can also enable an alternative structure that separates the metrics into an array.

Please reach out to techsupport@hawkindynamics.com to have any of these configuration options changed for your organization.

Methods

1. Get Access Token

Use the Refresh Token generated from <https://cloud.hawkindynamics.com> to get a valid Access Token. The token generator can be found in the settings menu in the upper right of the screen under the “Integrations” option. Only the organization administrator account has the ability to generate API tokens as the tokens have access to all data within the org.

Request

Method	URL
GET	https://cloud.hawkindynamics.com/api/token

Type	Params	Values
HEADER	Authorization	Bearer <refresh_token>

Example Request:

```
curl -H "Authorization: Bearer <refresh token>" https://cloud.hawkindynamics.com/api/token
```

Response

Status	Response
200	<pre>{ "access_token": <access_token>, "token_type": "Bearer", "expires_at": 1551301530, }</pre>
403	<pre>{"error": "Refresh Token is missing."}</pre>
401	<pre>{"error": "Refresh Token is invalid or expired."}</pre>
500	<pre>{"error": "Something went wrong. Please contact support@hawkindynamics.com"}</pre>

2. Get Tests

Get the tests for an account

Request

Method	URL
GET	<a href="https://cloud.hawkindynamics.com/api/<org_name>">https://cloud.hawkindynamics.com/api/<org_name>

Type	Params	Values
HEADER	Authorization	Bearer <access_token>
GET	from (optional)	Unix Timestamp
GET	to (optional)	Unix Timestamp
GET	syncFrom (optional)	Unix Timestamp
GET	syncTo (optional)	Unix Timestamp
GET	athleteId (optional)	Athlete ID
GET	teamId (optional)	Team ID or comma separated list of Team IDs (maximum of 10)
GET	groupId	Group ID or comma separated list of Group IDs (maximum of 10)
GET	testTypeId	Test Type ID

access_token

The `access_token` that was received in response to <https://cloud.hawkindynamics.com/api/token>

from

Optionally supply a time (Unix timestamp) you want the tests from. If you do not supply this value you will receive every test. This parameter is best suited for bulk exports of historical data.

to

Optionally supply a time (Unix timestamp) you want the tests to. If you do not supply this value you will receive every test from the beginning of time or the optionally supplied `from` parameter. This parameter is best suited for bulk exports of historical data.

syncFrom

Optionally supply a Unix timestamp to receive tests that have changed since that time. The result set will include updated and newly created tests. This parameter is best suited to keep your database in sync with the Hawkin database. It cannot and should not be used to fetch your

entire database. A recommended strategy would be to have a job that runs on a short interval e.g. every five minutes that sends the `lastSyncTime` that it received as the `syncFrom` parameter.

If you do not supply this value you will receive every test. This parameter is best suited for bulk exports of historical data.

syncTo

Optionally supply a time (Unix timestamp) to receive tests that have changed up until this time. This parameter is best suited to keep your database in sync with the Hawkin database.

athleteId

Optionally supply an athlete's id to receive tests for a specific athlete. Can only be used in conjunction with `from` and `to` parameters.

teamId

Optionally supply a team's or a comma separated list of team id's to receive tests from specific teams. A maximum of 10 teams can be fetched at once. Can only be used in conjunction with `from` and `to` parameters.

Ex.

https://cloud.hawkindynamics.com/api/<org_name>?teamId=teamOneId,teamTwoId,teamThreeId

groupId

Optionally supply a group's or a comma separated list of group id's to receive tests from specific groups. A maximum of 10 groups can be fetched at once. Can only be used in conjunction with `from` and `to` parameters.

Ex.

https://cloud.hawkindynamics.com/api/<org_name>?groupId=groupOneId,groupTwoId,groupThreeId

testTypeId

Optionally supply a test type id to only retrieve tests of that type. Can only be used in conjunction with `from` and `to` parameters.

NOTE:

Be aware that the available metrics can change at any time. Responses may include new metrics in the future and some tests may be missing metrics if they cannot be calculated.

Example Requests:

```
curl -H "Authorization: Bearer <access_token>" https://cloud.hawkindynamics.com/api/<org_name>
```

```
curl -H "Authorization: Bearer <access_token>"  
https://cloud.hawkindynamics.com/api/<org\_name>?from=1551301530
```

```
curl -H "Authorization: Bearer <access_token>"  
https://cloud.hawkindynamics.com/api/<org\_name>?from=1551301530&to=1552301530
```

Response

Status	Response
200	Response will be an array containing the tests that match this query Each test has the following structure: <pre>{ "id": <string>, "testType": { "id": <string>, "name": <string>, "canonicalId": <string>, "tags": [{ "id": <string>, "name": <string>, "description": <string>, }], }, "athlete": { "id": <string>, "name": <string>, }, "timestamp": <string>, # unix timestamp "segment": <string> #test type:trial in the session, "<MetricName>": <number string> #the metric value, }</pre>

Every test type has different metrics included.

An example response is:-

```
{
  "data": [
    {
      "id": "BMKtuEptQ1aIV0HE3oH15",
      "athlete": {
        "id": "BMKtuEptQ1aIV0HE3oH1",
        "name": "Athlete 1"
      },
      "segment": "Countermovement Jump:5"
      "testType": {
        "id": "9710f473daa20f7069c3d337753e879c",
        "name": "Countermovement Jump",
        "canonicalId": "7nNduHeM5zETPjHxvm7s",
        "tags": [{
          "id": "9710f473daa20f7069c3d337753e879c",
          "name": "Tag 1",
          "description": ""
        }],
      },
      "timestamp": 1551301560,
      "Relative Peak Landing Force(%)": 2,
      "System Weight(N)": 2,
      "L|R Peak Propulsive Force(%)": 1.5,
      "L|R Avg. Braking Force(%)": 1.5,
      "Peak Braking Power(W)": 2,
      "Impulse Ratio": 1.5,
      "Braking Phase %": 2,
      "Avg. Relative Propulsive Force(%)": 2,
      "Flight Time(s)": 1.5,
      "L|R Avg. Braking RFD(%)": 1.5,
      "Braking Phase(s)": 1.5,
      "Avg. Braking Velocity(m/s)": 1.5,
      "Peak Landing Force(N)": 2,
      "L|R Avg. Propulsive Force(%)": 1.5,
      "Peak Propulsive Force(N)": 2,
      "Peak Propulsive Power(W)": 2,
      "Avg. Braking Power(W)": 2,
      "Peak Velocity(m/s)": 1.5,
    }
  ]
}
```


	<pre> "Avg. Landing Force(N)": 2, "Landing Stiffness(N/m)": 2, "Unweighting Phase %": 2, "Countermovement Depth(m)": 1.5, "Time to Stabilization(ms)": "N/A", "Propulsive Net Impulse(N.s)": 2, "Braking Net Impulse(N.s)": 2, "Propulsive Phase(s)": 1.5, "Jump Height(m)": 1.5, "L R Landing Impulse Index(%)": "N/A", "L R Avg. Landing Force(%)": 1.5, "L R Braking Impulse Index(%)": 1.5, "Takeoff Velocity(m/s)": 1.5, "RSI": 1.5, "L R Peak Braking Force(%)": 1.5, "Propulsive Phase %": 2, "Peak Braking Force(N)": 2, }, "lastTestTime": 1551301560, "lastSyncTime": 1551301560, "count": 1, } </pre>
401	<pre>{"error": "Invalid Access Token."}</pre>
500	<pre> {"error": "Something went wrong. Please contact support@hawkindynamics.com"}</pre>

3. Get Athletes

Get the athletes for an account

Request

Method	URL
GET	<a href="https://cloud.hawkindynamics.com/api/<org_name>/athletes">https://cloud.hawkindynamics.com/api/<org_name>/athletes

Type	Params	Values
HEADER GET	Authorization includeInactive	Bearer <access_token> If present, the response will include inactive athletes

access_token

The `access_token` that was received in response to
<https://cloud.hawkindynamics.com/api/token>

Example Requests:

```
curl -H "Authorization: Bearer <access_token>"
https://cloud.hawkindynamics.com/api/<org_name>/athletes
```

Response

Status	Response
200	Response will be an array containing the athletes that match this query. Each athlete has the following structure: <pre>{ "id": <string>, "name": <string>, "active": <boolean>, "teams": <Array<string>>, #contains team ids "groups": <Array<string>>, #contains group ids "external?": { "<customProperty1>": "<customValue1>", "<customProperty2>": "<customValue2>", } }</pre> An example response is:- <pre>{ "data": [{</pre>

	<pre> "id": "id1", "name": "Athlete 1" }, { "id": "id2", "name": "Athlete 2" }, { "id": "id3", "name": "Athlete 3" }], "count": 3 } </pre>
401	<code>{"error": "Invalid Access Token."}</code>
500	<code>{"error": "Something went wrong. Please contact support@hawkindynamics.com"}</code>

4. Create Athletes

Bulk Create up to 500 Athletes at a time

Request

Method	URL
POST	<code>https://cloud.hawkindynamics.com/api/<org_name>/athletes/bulk</code>

Type	Params	Values
HEADER RequestBody	Authorization	<code>Bearer <access_token></code> <pre> [{ name: <string>, image?: <string>, #default=null active?: <boolean>, #default=true teams?: <Array<string>>, #default=[defaultTeamId] groups?: <Array<string>>, #default=[] external?: { </pre>

		<pre> customProperty1: <customValue1>, ... } }]</pre>
--	--	--

access_token

The `access_token` that was received in response to <https://cloud.hawkindynamics.com/api/token>

Response

Status	Response
200	Response will be an object. The <code>data</code> array will be the athletes that were successfully created with their unique id. <code>hasFailures</code> will be true if any athletes were not created, and the <code>failures</code> array will contain the reason why the athletes were not created <pre>{ data: [{ "id": <string>, "name": <string>, "active": <boolean>, "teams": <Array<string>>, #contains team ids "groups": <Array<string>>, #contains group ids "external?": { "<customProperty1>": "<customValue1>", "<customProperty2>": "<customValue2>", ... } }], "hasFailures": <boolean> "failures": [{ "reason": <string>, "data": { "id": <string>, "name": <string>, ... } }] }</pre>

401	{"error": "Invalid Access Token."}
413	{"error": "Payload Too Large"}
500	{"error": "Something went wrong. Please contact support@hawkindynamics.com"}

5. Create Athlete

Create an Athlete.

Request

Method	URL
POST	https://cloud.hawkindynamics.com/api/<org_name>/athletes

Type	Params	Values
HEADER RequestBody	Authorization	Bearer <access_token> <pre>{ name: <string>, #required image?: <string>, #default=null active?: <boolean>, #default=true teams?: <Array<string>>, #default=[defaultTeamId] groups?: <Array<string>>, #default=[] external?: { customProperty1: customValue1, } }</pre>

access_token

The `access_token` that was received in response to <https://cloud.hawkindynamics.com/api/token>

Response

Status	Response
200	Response will be an object containing the created athlete <pre> { "id": <string>, "name": <string>, "active": <boolean>, "teams": <Array<string>>, #contains team ids "groups": <Array<string>>, #contains group ids "external?": { "<customProperty1>": "<customValue1>", "<customProperty2>": "<customValue2>", ... } } </pre>
400	{"error": <string>}
401	{"error": "Invalid Access Token."}
500	{"error": "Something went wrong. Please contact support@hawkindynamics.com }

6. Update Athletes

Bulk Update up to 500 Athletes.

Request

Method	URL
PUT	https://cloud.hawkindynamics.com/api/<org_name>/athletes/bulk

Type	Params	Values
HEADER RequestBody	Authorization	<pre> Bearer <access_token> [{ id: <string>, #required name?: <string>, image?: <string>, active?: <boolean>, teams?: <Array<string>>, groups?: <Array<string>>, external?: { customProperty1:customValue1, ... } }]</pre>

* If optional fields are not present in an update request, those properties will be left unchanged. However, when updating external properties, custom properties that are not present will be removed.

access_token

The `access_token` that was received in response to <https://cloud.hawkindynamics.com/api/token>

Response

Status	Response
--------	----------

200	Response will be an object. The `data` array will be the athletes that were successfully updated. `hasFailures` will be true if any athletes were not updated, and the `failures` array will contain the reason why the athletes were not created <pre> { "data": [{ "id": <string>, "name": <string>, "active": <boolean>, "teams": <Array<string>>, #contains team ids "groups": <Array<string>>, #contains group ids "external?": { "<customProperty1>": "<customValue1>", "<customProperty2>": "<customValue2>", ... } }], "hasFailures": <boolean> "failures": [{ "reason": <string>, "data": { "id": <string>, "name": <string>, ... } }] } </pre>
401	{"error": "Invalid Access Token."}
413	{"error": "Payload Too Large"}
500	{"error": "Something went wrong. Please contact support@hawkindynamics.com"}

7. Update Athlete

Update an Athlete.

Request

Method	URL
PUT	<a href="https://cloud.hawkindynamics.com/api/<org_name>/athletes/<athleteId>">https://cloud.hawkindynamics.com/api/<org_name>/athletes/<athleteId>

Type	Params	Values
HEADER RequestBody	Authorization	Bearer <access_token> <pre>{ id: <string>, #required name?: <string>, image?: <string>, #default=null active?: <boolean>, #default=true teams?: <Array<string>>, #default=[defaultTeamId] groups?: <Array<string>>, #default=[] external?: { customProperty1: <customValue1>, ... } }</pre>

* If optional fields are not present in an update request, those properties will be left unchanged. However, when updating external properties, custom properties that are not present will be removed.

access_token

The `access_token` that was received in response to <https://cloud.hawkindynamics.com/api/token>

Response

Status	Response
200	Response will be an object containing the created athlete <pre>{ "id": <string>, "name": <string>, "active": <boolean>, "teams": <Array<string>>, #contains team ids "groups": <Array<string>>, #contains group ids "external?": { "<customProperty1>": <customValue1>, "<customProperty2>": <customValue2>, ... } }</pre>
400	<pre>{"error": <string>}</pre>
401	<pre>{"error": "Invalid Access Token."}</pre>
500	<pre>{"error": "Something went wrong. Please contact support@hawkindynamics.com"}</pre>

8. Get ForceTime Data

Get the force-time data for a specific test by id

Request

Method	URL
GET	<a href="https://cloud.hawkindynamics.com/api/<org_name>/forcetime/<test_id>">https://cloud.hawkindynamics.com/api/<org_name>/forcetime/<test_id>

Type	Params	Values
HEADER	Authorization	Bearer <access_token>

access_token

The `access_token` that was received in response to <https://cloud.hawkindynamics.com/api/token>

Example Requests:

```
curl -H "Authorization: Bearer <access_token>"  
https://cloud.hawkindynamics.com/api/<org\_name>/forcetime/<test\_id>
```

Response

Status	Response
200	Response will be an object containing the force-time data for the requested test. <pre>{ "id": <string>, "testType": { "id": <string>, "name": <string>, }, "athlete": { "id": <string>, "name": <string>, }, }</pre>

	<pre> "timestamp": <string>, #unix timestamp "Time(s)": [0.001,0.002,0.003,0.004...] "LeftForce(N)": [481,480,482,484...], "RightForce(N)": [481,480,482,484...], "CombinedForce(N)": [962,960,964,968...], "Velocity(m/s)": [0,0,0,0...], "Displacement(m)": [0,0,0,0...], "Power(W)": [0,0,0,0...], "rsi": "NA", </pre>
404	<pre>{"error": "Requested Resource Not Found."}</pre>
401	<pre>{"error": "Invalid Access Token."}</pre>
500	<pre>{"error": "Something went wrong. Please contact support@hawkindynamics.com"}</pre>

9. Get Teams

Get the team names and ids for all the teams in the org

Request

Method	URL
GET	<a href="https://cloud.hawkindynamics.com/api/<org_name>/teams">https://cloud.hawkindynamics.com/api/<org_name>/teams

Type	Params	Values
HEADER	Authorization	<code>Bearer <access_token></code>

access_token

The `access_token` that was received in response to <https://cloud.hawkindynamics.com/api/token>

Example Requests:

```
curl -H "Authorization: Bearer <access_token>"
```

https://cloud.hawkindynamics.com/api/<org_name>/teams

Response

Status	Response
200	Response will be an array containing the teams in the organization. Each team has the following structure: <pre>{ "id": <string>, "name": <string> }</pre>
401	<pre>{"error": "Invalid Access Token."}</pre>
500	<pre>{"error": "Something went wrong. Please contact support@hawkindynamics.com"}</pre>

10. Get Groups

Get the team names and ids for all the teams in the org

Request

Method	URL
GET	<a href="https://cloud.hawkindynamics.com/api/<org_name>/groups">https://cloud.hawkindynamics.com/api/<org_name>/groups

Type	Params	Values
HEADER	Authorization	Bearer <access_token>

access_token

The `access_token` that was received in response to <https://cloud.hawkindynamics.com/api/token>

Example Requests:

```
curl -H "Authorization: Bearer <access_token>"
```

https://cloud.hawkindynamics.com/api/<org_name>/groups

Response

Status	Response
200	Response will be an array containing the groups in the organization. Each group has the following structure: <pre>{ "id": <string>, "name": <string> }</pre>
401	<pre>{"error": "Invalid Access Token."}</pre>
500	<pre>{"error": "Something went wrong. Please contact support@hawkindynamics.com"}</pre>

11. Get Test Types

Get the test type names and ids for all the test types in the system

Request

Method	URL
GET	<a href="https://cloud.hawkindynamics.com/api/<org_name>/test_types">https://cloud.hawkindynamics.com/api/<org_name>/test_types

Type	Params	Values
HEADER	Authorization	<code>Bearer <access_token></code>

access_token

The `access_token` that was received in response to <https://cloud.hawkindynamics.com/api/token>

Example Requests:

```
curl -H "Authorization: Bearer <access_token>"
```

https://cloud.hawkindynamics.com/api/<org_name>/test_types

Response

Status	Response
200	Response will be an array containing the test type. Each test type has the following structure: <pre>{ "id": <string>, "name": <string> }</pre>
401	<pre>{"error": "Invalid Access Token."}</pre>
500	<pre>{"error": "Something went wrong. Please contact support@hawkindynamics.com"}</pre>

12. Get Metrics

Get the metrics and ids for all the metrics in the system

Request

Method	URL
GET	<a href="https://cloud.hawkindynamics.com/api/<org_name>/metrics">https://cloud.hawkindynamics.com/api/<org_name>/metrics

Type	Params	Values
HEADER	Authorization	Bearer <access_token>

access_token

The `access_token` that was received in response to <https://cloud.hawkindynamics.com/api/token>

Example Requests:

```
curl -H "Authorization: Bearer <access_token>"
```

https://cloud.hawkindynamics.com/api/<org_name>/metrics

Response

Status	Response
200	Response will be an array containing each test type and the metrics for that test type. Each test type has the following structure: <pre>[{ "canonicalTestId": <string>, "testType": <string>, "metrics": [{ "id" : <string>, "label": <string>, "units": <string null>, "description": <string> },], },]</pre>
401	<pre>{"error": "Invalid Access Token."}</pre>
500	<pre>{"error": "Something went wrong. Please contact support@hawkindynamics.com"}</pre>

13. Get Tags

Get the tag names and ids for all the tags in the system

Request

Method	URL
GET	<a href="https://cloud.hawkindynamics.com/api/<org_name>/tags">https://cloud.hawkindynamics.com/api/<org_name>/tags

Type	Params	Values
------	--------	--------

HEADER	Authorization	Bearer <access_token>
--------	---------------	-----------------------

access_token

The `access_token` that was received in response to <https://cloud.hawkindynamics.com/api/token>

Example Requests:

```
curl -H "Authorization: Bearer <access_token>"
https://cloud.hawkindynamics.com/api/<org_name>/tags
```

Response

Status	Response
200	Response will be an array containing the tags. Each tag has the following structure: <pre>{ "id": <string>, "name": <string>, "description": <string>, }</pre>
401	<pre>{"error": "Invalid Access Token."}</pre>
500	<pre>{"error": "Something went wrong. Please contact support@hawkindynamics.com"}</pre>